Dynamic Predictive Model for Growth of *Salmonella* Enteritidis in Egg Yolk

V. GUMUDAVELLI, J. SUBBIAH, H. THIPPAREDDI, P.R. VELUGOTI, AND G. FRONING

ABSTRACT: *Salmonella* Enteritidis (SE) contamination of poultry eggs is a major human health concern worldwide. The risk of SE from shell eggs can be significantly reduced through rapid cooling of eggs after they are laid and their storage under safe temperature conditions. Predictive models for the growth of SE in egg yolk under varying ambient temperature conditions (dynamic) were developed. The growth of SE in egg yolk under several isothermal conditions (10, 15, 20, 25, 30, 35, 37, 39, 41, and 43 °C) was determined. Baranyi model, a primary model, was fitted with growth data for each temperature and corresponding maximum specific growth rates were estimated. Root mean squared error (RMSE) values were less than 0.44 $\log_{10}$ CFU/g and pseudo-R^2 values were greater than 0.98 for the primary model fitting. For developing the secondary model, the estimated maximum specific growth rates were then modeled as a function of temperature using the modified Ratkowsky's equation. The RMSE and pseudo-R^2 were 0.05/h and 0.99, respectively. A dynamic model was developed by integrating the primary and secondary models and solving it numerically using 4th-order Runge–Kutta method to predict the growth of SE in egg yolk under varying temperature conditions. The integrated dynamic model was then validated with 4 temperature profiles (varying) such as linear heating, exponential heating, exponential cooling, and sinusoidal temperatures. The predicted values agreed well with the observed growth data with RMSE values less than 0.29 $\log_{10}$ CFU/g. The developed dynamic model can predict the growth SE in egg yolk under varying temperature profiles.

Keywords: egg yolk, lactates, predictive model, *Salmonella* Enteritidis, temperature

Introduction

Salmonellosis continues to be a major food-borne disease despite efforts by the food industry. The prevalence of *Salmonella* Enteritidis (SE) is greater in chicken and in shell eggs (CDC 2006). Of the total number of eggs produced in 2005, 59% were marketed as shell eggs. Shell eggs can be contaminated with SE as a result of infected laying hens (Rodrigue and others 1990). While SE growth can be minimal in the albumen, SE can contaminate the egg yolk if SE penetrates the vitelline membrane during storage and grows rapidly in iron-rich yolk after transmission through the egg shell and the albumen (Humphrey and others 1991; Humphrey and Whitehead 1993). The growth of SE depends on the egg cooling rate and the temperature of storage of the shell eggs (Humphrey 1994).

The research literature on the growth of SE in eggs indicates that the studies are restricted to growth of SE only at very few selected temperatures. Some of the notable studies include growth and penetration of SE in shell eggs at 25 °C (Grijspeerdt 2001); the effect of immunization of hens on the growth of SE in yolk (Gurtler and Fehlhaber 2004); the growth and penetration of SE at temperatures of 4, 10, and 25 °C (Schoeni 1995), and the influence of the initial inoculum level on SE growth at 10, 17.5, and 25 °C (Gast and Holt 2000). In the present investigation, the growth of SE in egg yolk was studied because its growth in egg albumen will be minimal due to the presence of antimicrobial lysozyme (Grijspeerdt and others 2005).

Predictive microbiology offers various tools in the form of mathematical models that can be handy to determine the growth of SE in shell eggs during storage. Temperature is the major factor affecting microbial growth and temperature can change drastically during processing and storage. Therefore, the effect of temperature on the growth rate of SE was studied in the present investigation. Predictive microbial models can be classified into primary and secondary models. While a primary model describes the growth of bacteria over a time period under constant environmental conditions such as temperature, the secondary model describes the effect of environmental conditions on the parameters of the primary model (Whiting 1995). A dynamic model integrates the primary and the secondary models to predict the growth of bacteria under varying environmental conditions.

Several types of primary models such as the modified Gompertz and the logistic can be used to fit the growth of microorganisms against time. When these models were evaluated for dynamic conditions, an incremental value has to be added to the initial inoculum to solve the differential equations (Huang 2003) that can be defined in a way to match the predicted and observed values. It also has an influence on the lag phase of the microbial growth curve. Therefore, these sigmoidal-growth models are excellent to describe the growth at constant temperatures, but not suitable for fluctuating temperatures.

The Baranyi model was chosen in this study because it has been applied by several researchers to successfully predict the microbial growth under fluctuating temperatures (Baranyi and others 1995; McClure and others 1997; Sutherland and others 1997; Pin and others 1999). A complete study of development and validation of primary, secondary, and dynamic models was conducted and Q1

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to predict growth of *Salmonella* Typhimurium on sterile chicken where the experimental data were collected from 10 to 40 °C (Oscar 2005).

The USDA-FSIS risk assessments of SE in shell eggs and *Salmonella* spp. in egg products stated that there is sparse data available for estimating growth parameters for SE and complete growth data suitable for predictive models were totally absent (FSIS 2005). The objective of this study was to develop and validate a predictive model of SE growth in egg yolk under fluctuating temperatures for use in evaluating the risk of SE growth during cooling of shell eggs after they were laid and during processing, storage, and distribution.

Materials and Methods

Organisms

Five SE strains (MG03835, MG04498, MG07928, MB00617, MB00629, all isolates from egg-associated outbreaks) were used. The strains were preserved individually on cryogenic beads (CryoBank; Copan Diagnostics, Corona, Calif., U.S.A.). Fresh SE cultures were prepared by inoculating 0.1 mL from the thawed stock culture into 10 mL of Tryptic Soy Broth (TSB; Difco, Detroit, Mich., U.S.A.) and incubation for 18 to 21 h at 35 °C. A cocktail containing all 5 strains of SE serotypes was prepared immediately before inoculation by mixing 5 mL of each strain in a centrifuge tube. The contents were centrifuged at 7,012 × *g*, where *g* is acceleration due to gravity, for 15 min at 4 °C. The supernatant was discarded and the bacterial cell pellet was reconstituted in 9 mL of 0.1% sterile peptone water. Cell suspension was adjusted to an absorbance of 1 to 1.10 at 600 nm using a spectrophotometer (Beckman, Model Q2 DU-65) so that the SE concentration was between 2 and 3 $\log_{10}$ CFU/g.

Preparation and inoculation of egg yolk

Fresh shell eggs (laid on same day or on previous day) were collected from poultry farm at the Univ. of Nebraska, Lincoln. Large, Grade A shell eggs were washed and sanitized to remove external contamination such as fecal matter, dust, and so on. Shell eggs were aseptically broken with an alcohol sanitized knife (70% ethanol) and yolks were separated from albumens. The yolks were then collected in a sterile beaker.

Appropriate volume of the 5-strain SE cocktail was added to the egg yolk in a stomacher bag (Spiral Biotech, Norwood, Mass., U.S.A.) to attain a concentration of approximately 2.5 $\log_{10}$ CFU/g. The inoculated yolk was stomached for 30 s. Five gram portions of the inoculated egg yolk were transferred to vacuum pouches (6.35 × 12.7 cm; Prime Source, Kansas City Mo., U.S.A.; 3 mil standard barrier nylon pouch, oxygen transmission rate of 3000 $cm^3/m^2/24$ h at 23 °C and 1 atm) and were heat sealed.

Isothermal growth data

Growth data of SE at 7, 10, 15, 20, 25, 30, 35, 37, 39, 41, 43, and 45 °C were collected to cover the entire bio-kinetic growth temperatures. The inoculated samples (pouches) were immersed in water baths maintained at constant temperatures. Two independent replications for each temperature were performed. Twelve pouches were prepared for each replicate. About 10 min were allowed for temperature stabilization, and a pouch was removed to estimate initial inoculum using microbial enumeration. Remaining pouches were removed at specific intervals until a stationary phase of SE was attained at each isothermal condition.

Dynamic growth data

Four different temperature profiles, linear cooling (from 43 to 7 °C for 12 h), exponential cooling (from 43 to 7 °C for 7 h), exponential heating (from 7 to 43 °C for 11 h), and sinusoidal heating and cooling (between 3 and 43 °C for 24 h) were used to evaluate SE growth in egg yolk under constantly changing temperature conditions. Programmable water baths with water circulation capabilities (RTE 740, Thermo Neslab, Portsmouth, N.H., U.S.A.) were used to run the temperature profiles. A commercial software package (NesCom Software, Portsmouth, N.H., U.S.A.) was used to program the temperature profiles in the water baths. In all, 15 samples for each temperature profiles were analyzed for SE enumeration.

SE enumeration

The pouches were removed from the water baths at predetermined intervals, wiped with paper towel, sprayed with alcohol, aseptically opened, and the yolk was transferred to a filter stomacher bag. Peptone water (20 mL) was added to the yolk sample and homogenized in a stomacher (Stomacher 400, Seward Medical, London, U.K.) for 1 min. Serial dilutions were prepared using peptone water and 1 mL of sample was either spread or spiral plated on xylose lysine deoxycholate (XLD) agar (Difco), and the plates were incubated for 48 h at 35 °C. Colonies were manually counted and microbial populations were then estimated.

Models

Primary model. Baranyi and Roberts (1994) proposed a dynamic model in a set of differential equations (Eq. 1a and 1b) to predict the growth of microbe under time-varying temperature conditions.

$$\frac{dy}{dt} = \frac{1}{1+e^{-Q}}\mu_{\max}\left(1-e^{(y-y_{\max})}\right) \tag{1a}$$

$$\frac{dQ}{dt} = \mu_{\max} \tag{1b}$$

with initial conditions $y = y_0$ at $t = 0$ and $Q = \log_e(q_0)$ where q is a measure of the initial physiological state of cells. At constant temperature, the 2 equations above can be analytically solved and the resulting explicit solution (Eq. 1c) is a 4-parameter model. The 4-parameter Baranyi model is widely used to describe the bacterial growth at static temperature conditions (Baranyi and Roberts 1994):

$$y(t) = y_0 + \mu_{\max}F(t) - \log_e\left(1+\frac{e^{\mu_{\max}F(t)}-1}{e^{(y_{\max}-y_0)}}\right) \tag{1c}$$

where $F(t) = t + \frac{1}{\nu}\log_e\left(e^{-\nu t}+e^{-h_0}-e^{(-\nu t-h_0)}\right)$; $y(t)$ is the $\log_e$CFU/g of cell concentration at time t; y_0 is initial cell concentration in $\log_e$CFU/g; $y_{\max}$ is maximum cell concentration in $\log_e$CFU/g; $\mu_{\max}$ is maximum specific growth rate in terms of $\log_e$CFU/g in 1/h; ν is rate of increase of the limiting substrate, assumed to be equal to $\mu_{\max}$; and h_0 is equal to $\mu_{\max *}\lambda$; λ is lag-phase duration in hours. Fitting of the growth data and analysis of results were all done in $\log_e$CFU/g units. However, results were reported in $\log_{10}$ CFU/g units for simplicity purposes.

Baranyi and Roberts (1994) suggested that in situations where preinoculation history of the cells is identical, the value of h_0 is approximately constant. Therefore, observed data was fitted to the Baranyi model and the parameters y_0, $y_{\max}$, $\mu_{\max}$, and h_0 were estimated at each temperature. The parameter h_0 was fixed at the average value of h_0 estimated for all the temperatures and the observed

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data were fitted again to the model and the other 3 parameters were estimated.

Doubling times of SE were calculated at each static temperature by $\log_e(2)$ divided by the maximum specific growth rate. Lag time was also calculated at each static temperature condition by dividing the h_0 value with the maximum specific growth rate estimated by the primary model fitting (Amézquita 2005).

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Secondary model. The modified Ratkowsky equation was used to describe the effect of temperature on growth rate, as given by (Zwietering and others 1991):

$$\mu_{\max} = a(T - T_{\min})^2 \left(1 - \exp(b(T - T_{\max}))\right) \tag{2}$$

where the parameters $T_{\min}$ and $T_{\max}$ represent theoretical minimum and maximum temperatures beyond which growth of the organism is impossible. The parameters a and b are regression coefficients. The 1st term of Eq. (2) represents the growth rate up to the optimum growth while the 2nd term describes the decrease in growth rate, as the temperature goes beyond the optimal growth temperature.

Modeling

Primary model fitting. A nonlinear regression procedure, PROC NLIN, in the SAS package (SAS 2006) was used to fit the data to the models with the Marquardt iterative method for estimation of parameters. Defining starting values to estimate the parameters in the SAS program is very critical because use of inappropriate values can lead to over or underestimation of the parameters. A Cartesian plot of the microbial count in $\log_{10}$ CFU/g units versus time in hours at each temperature was used to calculate the starting values for the parameters y_0 and $y_{\max}$. For each temperature, the value of the lower asymptote was given as a starting value for y_0 and upper asymptote for $y_{\max}$. The starting value of the maximum growth rate was the slope of the straight-line region the or exponential phase of the sigmoidal curve between the lag phase and the stationary phase. The starting value for h_0 was calculated by dividing the maximum growth rate by the lag time. The lag time is the corresponding value on the abscissa from the origin to the point where the exponential phase started. The given starting values, after several iterations in the nonlinear procedure, converged to estimated values of the parameters.

Unlike linear regression, the R^2 is not valid for nonlinear regression. Therefore, another measure called the pseudo-R^2 (Eq. 3) defined by Schabenberger (2005) was used to evaluate the goodness of fit.

$$\text{Pseudo} - R^2 = 1 - \frac{\text{SSR}}{\text{TSS}} \tag{3}$$

where SSR is sum of squares of residuals and TSS is total sum of squares.

Root mean squared error (RMSE) (Eq. 4) was also used to evaluate the model performance.

$$\text{RMSE} = \sqrt{\frac{(O - P)^2}{N - p}} \tag{4}$$

where O and P are observed and predicted microbial populations in $\log_{10}$ CFU/g, N is the number of observation, and p is the number of model parameters. The above 2 statistics were calculated for each temperature after fitting the observed growth data into the Baranyi model.

Secondary model fitting. The maximum specific growth rate estimated from each primary model at various temperatures was then fitted into the modified Ratkowsky 4 parameter equation (Eq. 2). The extreme temperatures where the SE growth was obtained (7 and 45 °C) were designated as starting values to the parameters $T_{\min}$ and $T_{\max}$. The parameters a and b were regression coefficients. The parameter a was determined by calculating the slope when $\sqrt{\mu_{\max}}$ values were plotted against T until the optimum growth rate occurred (Eq. 5).

$$\sqrt{\mu_{\max}} = a(T - T_{\min}) \tag{5}$$

RMSE and pseudo-R^2 were used as criteria to evaluate the performance of the secondary models.

Dynamic model

A computer simulation algorithm was developed to predict SE growth under fluctuating temperatures. The secondary model (Eq. 2) was incorporated into the differential form of the Baranyi model (Eq. 1a and 1b) and then solved numerically using the 4th-order Runge–Kutta method by using the MATLAB 7.0 package to predict the microbial population at each time step of 60 s. Various statistical criteria such as mean relative percentage error (MRPE) (Eq. 6), bias factor (BF) (Eq. 7), mean absolute relative error (MARE) (Eq. 8), accuracy factor (AF) (Eq. 9), RMSE (Eq. 5), and the correlation coefficient (r) were used to compare the observed and predicted values (Jeyamkondan and others 2001).

$$\text{MRPE} = \frac{1}{N}\sum \frac{(O - P)}{O} \times 100 \tag{6}$$

$$\text{BF} = 10^{\sum \log(P/O)/N} \tag{7}$$

$$\text{MARE} = \frac{1}{N}\sum \left|\frac{(O - P)}{O}\right| \times 100 \tag{8}$$

and

$$\text{AF} = 10^{\sum |\log(P/O)|/N} \tag{9}$$

An MRPE value close to zero is desired, and indicates no bias in prediction. A positive value of MRPE suggests the model underpredicted (fail-dangerous region) and a negative value indicates overprediction (fail-safe region). A BF value greater than 1 indicates the model overpredicted and *vice versa*. The MARE is similar to the MRPE except that the MARE takes into account the absolute values; therefore, a least positive value is desired. Both the MARE and the MRPE are expressed as percentage values. The AF measures the overall model error. An AF value of 1.1 indicates the average deviation of the predicted values from the observed values is 10%.

Pathogen modeling program (PMP), a predictive microbiology program, developed by the Eastern Regional Research Center, USDA can predict the growth and survival of several food-borne pathogens under combinations of several constant environmental conditions (PMP 2006). Most of these models are based on the growth parameters derived from model microbiological media, rather than the actual food matrix. Thus, some of the predictions from this application can be erroneous and also the PMP does not have models for SE growth. Other major constraints with PMP are that the microbial growth can be predicted at temperatures in the range of 10 to 30 °C, with an initial minimum inoculum greater than

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3 $\log_{10}$ CFU/mL. In addition, the user has to select a 'lag' and 'no-lag phase' options. Therefore, a comparison was performed on predictions by the PMP and the model developed in this study.

Results and Discussion

Primary model

SE growth was not observed in egg yolk at 7 °C (Figure 1) over a period of 40 d, while the growth was erratic at 45 °C (Figure 2) over a 12-h period. Hence, the primary model could not be fitted to the data at these temperatures. Stationary phase was attained within 625, 175, 50, 24, 21, 14, 11, 10, 9, and 11 h at 10, 15, 20, 25, 30, 35, 37, 39, 41, and 43 °C, respectively, for SE in egg yolk (Figure 3). The time interval at which microbial enumerations were performed for replicate 1 was different from that for replicate 2, as the observed growth data on Replicate 1 was used to identify appropriate time interval for microbial enumeration for Replicate 2. Care was taken to maintain an initial inoculum level of 2.5 $\log_{10}$ CFU/g between the 2 replications at all temperatures, but a small difference in these values at 15, 30, and 41 °C was observed. The RMSE values calculated after fitting the Baranyi model at various temperatures ranged from 0.13 to 0.44 $\log_{10}$ CFU/g with a mean value of 0.25 $\log_{10}$ CFU/g. The pseudo-R^2 values calculated at any temperature were all greater than 0.97.

The average y_{max} value estimated at all static temperatures was 8.65 $\log_{10}$ CFU/g. The h_0 values were estimated for all temperatures by fitting the respective growth data at all temperatures and they ranged from 0.72 to 1.93. The average value of h_0 was found out to be 1.34. This value was fixed for h_0 and then the Baranyi model was refitted to the growth data at all temperatures estimating the rest of the 3 parameters, y_0, y_{max}, and μ_{max}. Amézquita and others (2005) used a similar approach for developing growth model of *Clostridium perfringens* in cooked boneless ham. The reported mean h_0 value for *C. perfringens* in cooked, boneless ham was 6.7, which is

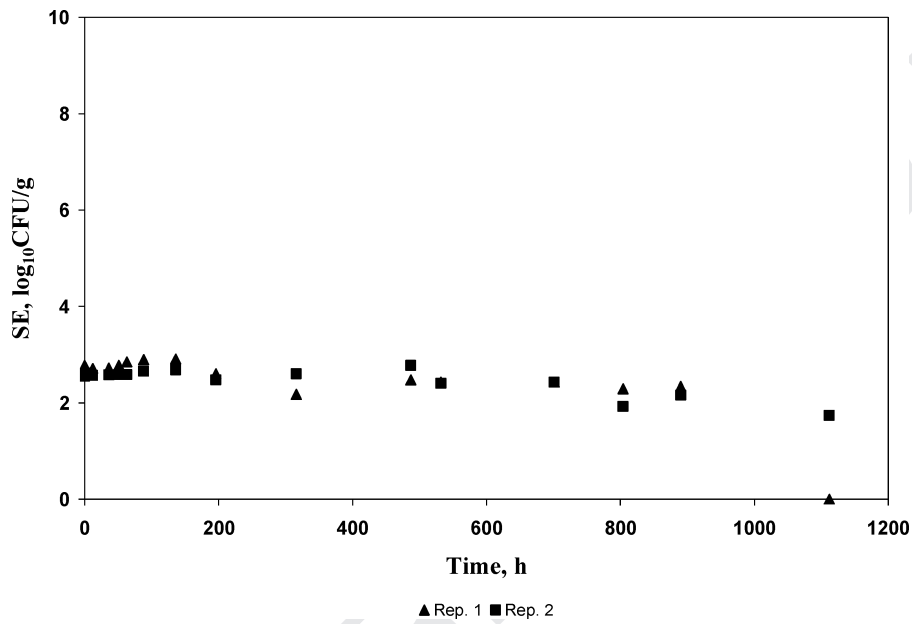

Figure 1—Population dynamics of SE in egg yolk at 7 °C

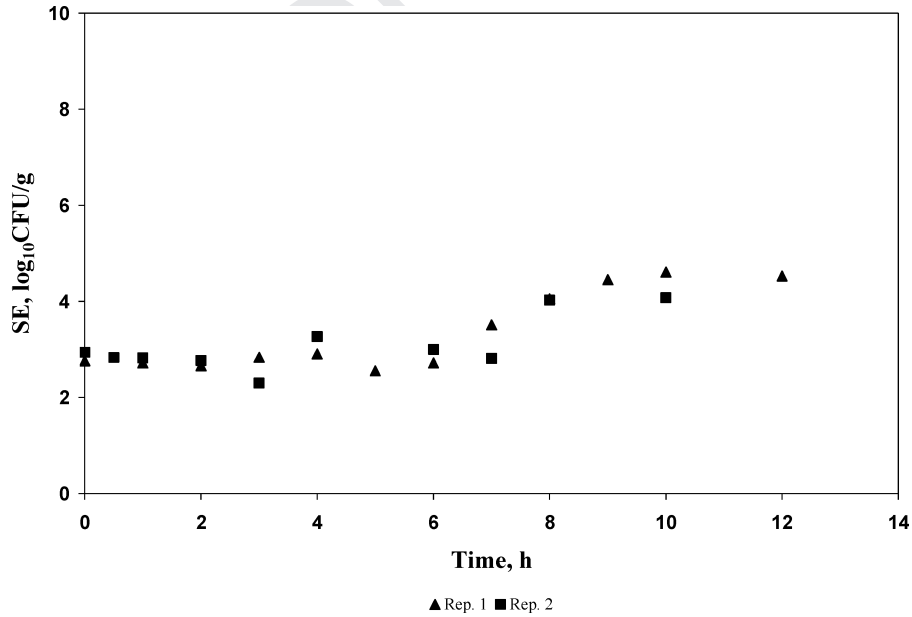

Figure 2—Population dynamics of SE in egg yolk at 45 °C

higher than the value from this study. This may be due to the difference in the organism used and also the food substrate used. When the yolk was inoculated with 15 and 150 CFU/mL, maximal SE populations of 8.4 and 8.7 $\log_{10}$ CFU/mL were attained, respectively, after 3 d at 25 °C, 8.0 and 8.2 $\log_{10}$ CFU/mL at 17.5 °C and only up to 2.8 and 3.1 $\log_{10}$ CFU/mL at 10 °C (Schoeni, 1995; Gast & Holt, 2000). These values agree with those observed in the current study (Figure 3). At 10 °C, SE grew up to 3 and 8 $\log_{10}$ CFU/mL at 3 and 25 d, respectively (Figure 3).

Secondary model

The secondary model for SE in shell egg yolk is given by Eq. 10.

$$\mu_{\max} = 0.002065(T - 6.13)^2.(1 - \exp(0.27469(T - 46.26))) \quad (10)$$

The RMSE and pseudo-R^2 values were 0.05/h and 0.99, respectively, for the secondary model fitting (Figure 4). The theoretical minimum ($T_{\min}$) and maximum ($T_{\max}$) temperatures at which SE was capable of growth in shell eggs were estimated as 6.13 and 46.26 °C, respectively, although SE growth was not observed from experimental data at 7 and 45 °C. This is due to the fact that $T_{\min}$ and $T_{\max}$ are model parameters and are usually 5 to 10 °C below or above the observed minimum and maximum temperatures of growth, respectively (Ross and Dalgaard 2004).

Dynamic model

Primary and secondary model predictions can be compared for dependent data while dynamic model can only be verified if the model was properly constructed (Oscar 2005). Therefore, for validating the dynamic model, data was collected independently without using any data that were used for primary model fitting. Further, the dynamic model was validated by collecting microbial data at fluctuating temperature profiles (Figure 5). The input to the dynamic model was y_0, the initial inoculum, and time–temperature profile. The value of $y_{\max}$ was fixed at 8.66 $\log_{10}$ CFU/g and h_0 was fixed at 1.34.

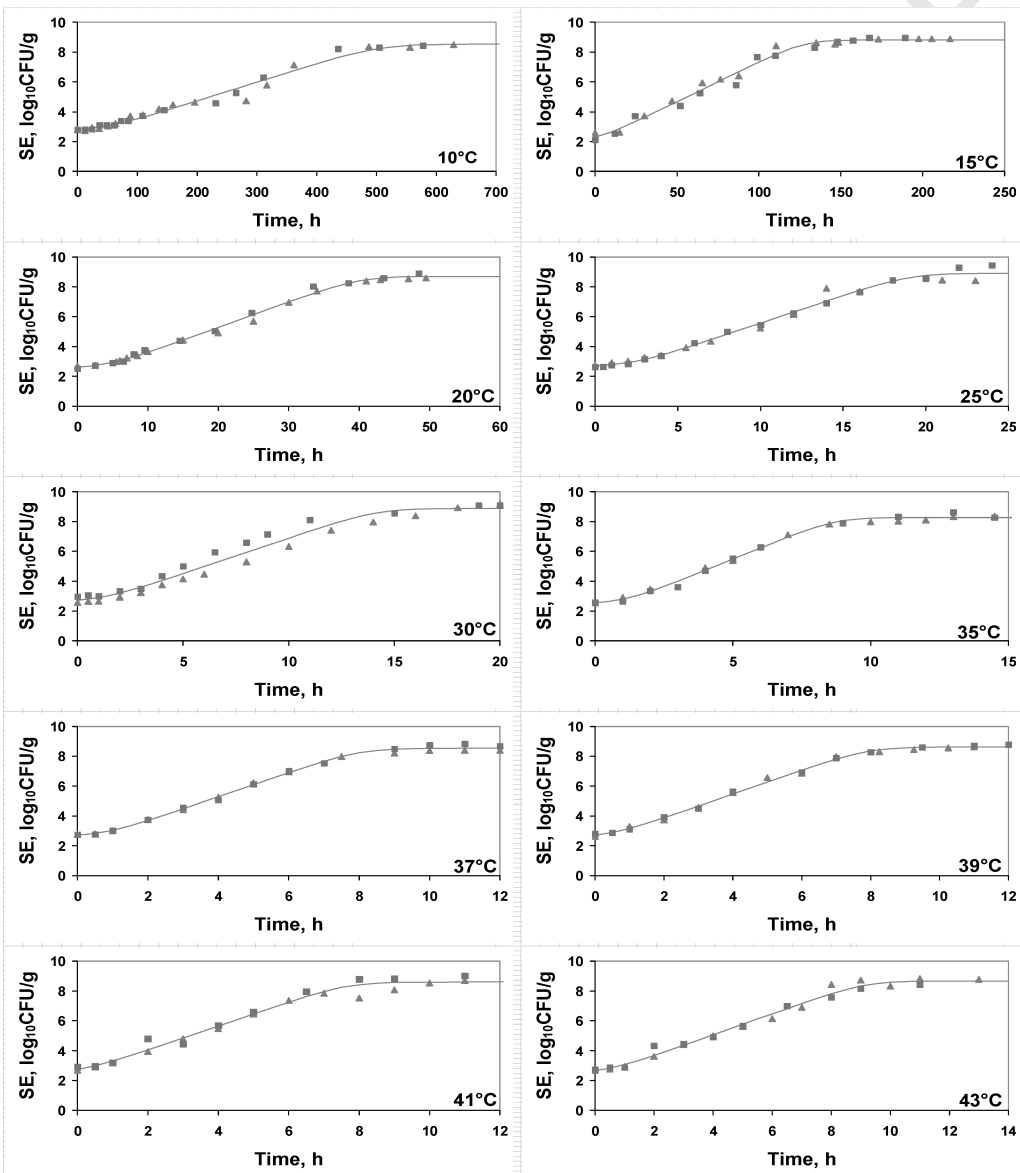

Figure 3—Fitting of growth data at different temperatures to the Baranyi model.

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Validation

For the validation of the exponential cooling profile (Figure 5a), the model slightly underpredicted the growth; therefore, there are some data points lying in the fail-dangerous region though close to the diagonal. For this reason, the MRPE value for this profile was higher (Table 1). A similar observation can be made for validation of the exponential heating profile that has the highest MRPE value (Figure 5d). The points were equally distributed into fail-safe and fail-dangerous regions for the profiles, linear cooling (Figure 5b) and sinusoidal (Figure 5c). The maximum value of RMSE for sinusoidal profile from Table 1 can be evidenced by an outlier at the 8-h point (Figure 5e). The correlation coefficient values between observed and predicted microbial populations were all greater than 0.99 for all validation profiles, indicating that the model performed satisfactorily (Table 1). Linear cooling and sinusoidal had the BF of 1, indicating that there was no bias. However, the BF values were less than 1 for exponential cooling and heating, indicating that the model underpredicted the microbial population. This is fail-dangerous. The MRPE values are in accordance with BF values. MRPE values for linear cooling and sinusoidal profiles were close to zero, indicating there was a negligible bias. Exponential heating and cooling profiles had positive values, indicating there was 2% and 2.96% underprediction, respectively. Overall, the RMSE values for all the profiles were all lower than 0.29 $\log_{10}$ CFU/g. Traditional microbial plating methods have an imprecision of 0.5 $\log_{10}$ CFU/g.

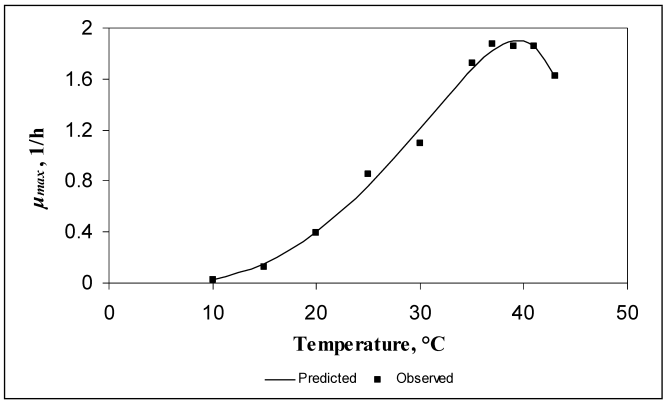

Figure 4—Secondary model to describe the growth rate of SE in egg yolk at various temperatures

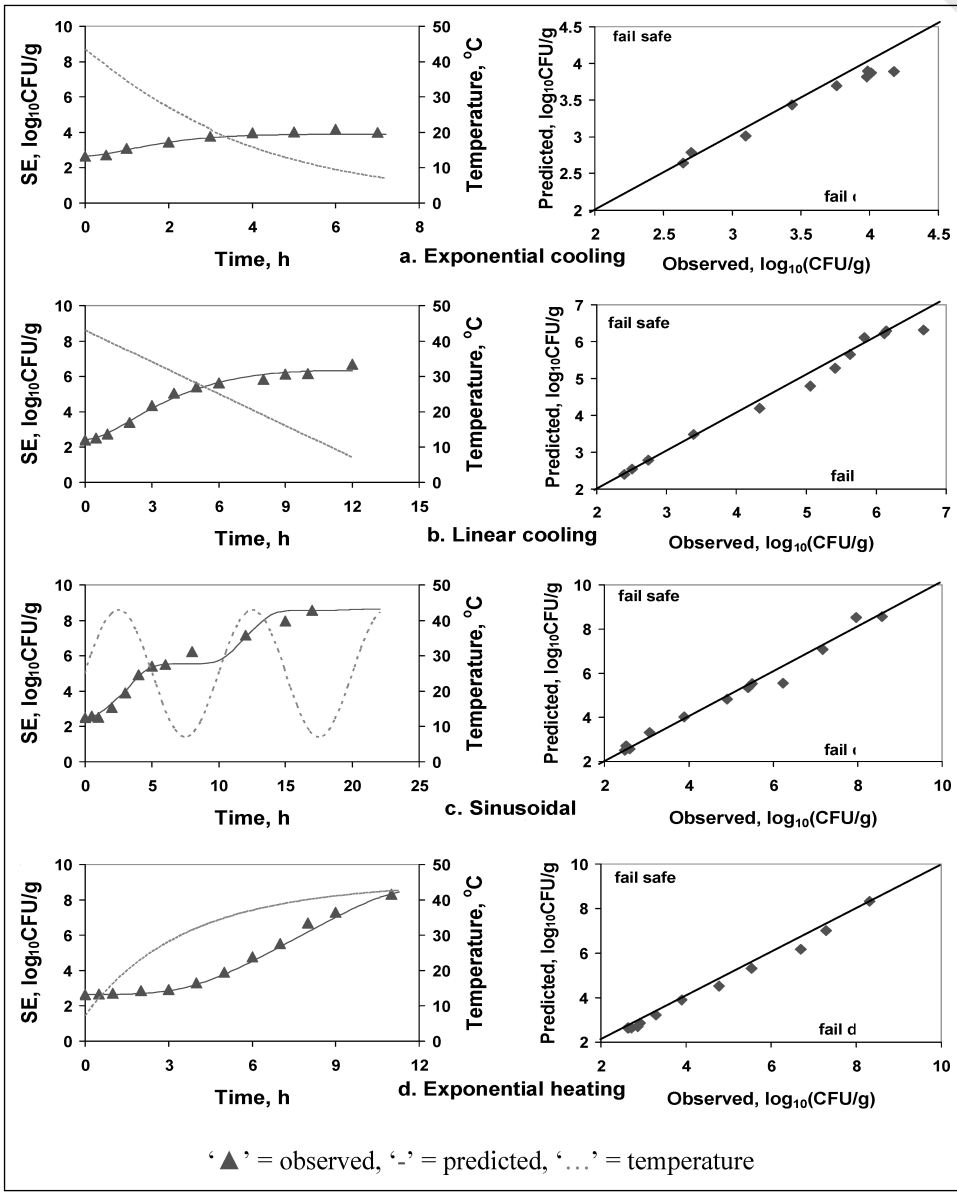

Figure 5—Validation of the predictive model at different fluctuating temperature profiles

RMSE values were of the order of the imprecision of reference microbial plating methods, and therefore that the model predicted the growth data well.

It is important to note that a cocktail of 5 strains was used in this study. The collected data and developed model predict the growth of the fastest growing strain in the cocktail. It is possible that a single strain of unknown relative growth potential may contaminate the egg yolk and may probably grow slower than what the model prediction. In such case, our model is conservative and overpredicts the growth. Therefore, the goodness-of-fit measures of our model in a real life situation will not be as good as what were reported in this study, because of the variation between strains.

Doubling times and lag times calculated at the temperatures where the growth data was obtained is shown in Table 2. The doubling time is the amount of time required for a single cell to multiply into 2. Table 2 shows, unlike maximum specific growth rates, the doubling time was greater at lower temperatures and as the temperature increased the doubling time decreased up to the optimum growth level. The doubling time again started increasing beyond the optimum growth temperatures. A 3 $\log_{10}$ CFU/g increase in SE would occur in 10 times of doubling time. Thus, a single cell of SE can grow to 3 $\log_{10}$ CFU/g in 43 d at a low temperature like 10 °C and in 4 h at optimum growth temperature, that is, 39 °C. Therefore, storage of eggs at temperatures ≤7.2 °C will minimize the potential for SE growth and reduce the risk of salmonellosis from eggs. Even at abusive refrigerated temperatures (10 °C), it takes several days for SE to grow to dangerous levels. However, if eggs are stored at room temperatures, it takes only a few hours for SE to grow to dangerous levels. Our data has shown that SE growth in egg yolk was restricted at 7 °C (Figure 1), validating the USDA-FSIS recommendation that shell eggs be cooled and stored ≤7.2 °C (45 °F) to assure their safety. Therefore, eggs must be cooled rapidly and refrigerated at all times.

Table 1—Comparison of various statistical criteria for validation of the predictive model at different temperature profiles

Profile	MRPE (%)	BF	MARE (%)	AF	RMSE ($\log_{10}$ CFU /g)	r
Exponential cooling	2.00	0.98	2.70	1.03	0.13	0.99
Linear cooling	0.17	1.00	2.60	1.03	0.17	0.99
Sinusoidal	−0.77	1.00	3.58	1.04	0.28	0.99
Exponential heating	2.96	0.97	2.99	1.03	0.21	0.99

Table 2—Doubling times and lag phase duration of SE in egg yolk

Temperature, °C	Doubling time, h	Lag time, h
10	32.84	47.39
15	4.57	5.84
20	1.73	4.46
25	0.91	2.26
30	0.57	1.25
35	0.41	1.10
37	0.38	0.83
39	0.36	0.58
41	0.37	0.39
43	0.43	0.59

As the lag time is inversely proportional to the maximum specific growth rate, the calculated lag times were greater at lower temperatures unlike the maximum specific growth rates. At lower temperatures such as 10 °C, the SE required more than 48 h to adapt to the new environment. As the growth conditions became favorable at higher temperatures, the lag time decreased and was least (24 min) at optimum growth temperature (Table 2).

PMP models predict the growth of *Salmonella* spp. in broth solution (PMP 2006) and it is of interest to compare the model performance for SE growth in egg yolk. The predictions from PMP were obtained by defining a pH of 6.7 and NaCl salt concentration of 0.5% g/dL to achieve a water activity of 0.997. Also, the option of 'lag phase' was selected over 'no-lag phase' for the prediction. SE growth using the developed dynamic model to those from PMP at 5 temperatures: 10 (Figure 6), 15, 20, 25, and 30 °C (Figure 7) was plotted. The observed data and the predictions by dynamic model at these temperatures were plotted on the same chart for comparison. However, PMP model requires the initial inoculum to be at least 3.0 $\log_{10}$ CFU/g. For comparison purposes, the predictions by our dynamic model with initial inoculum of 3.0 $\log_{10}$ CFU/g were also plotted. The PMP overpredicted SE growth at lower temperatures 10 (Figure 9), 15, and 20 °C and showed similar growth with higher temperatures (Figure 10). If a 'no-lag phase' option had been selected for PMP, the model would have been more conservative,

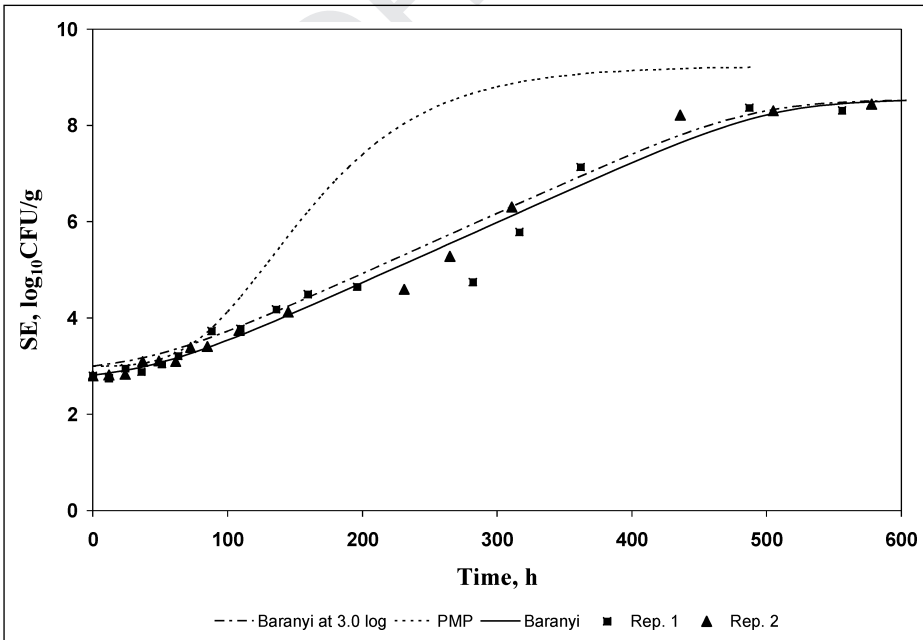

Figure 6—Comparison of the Baranyi model with the PMP model at 10 °C

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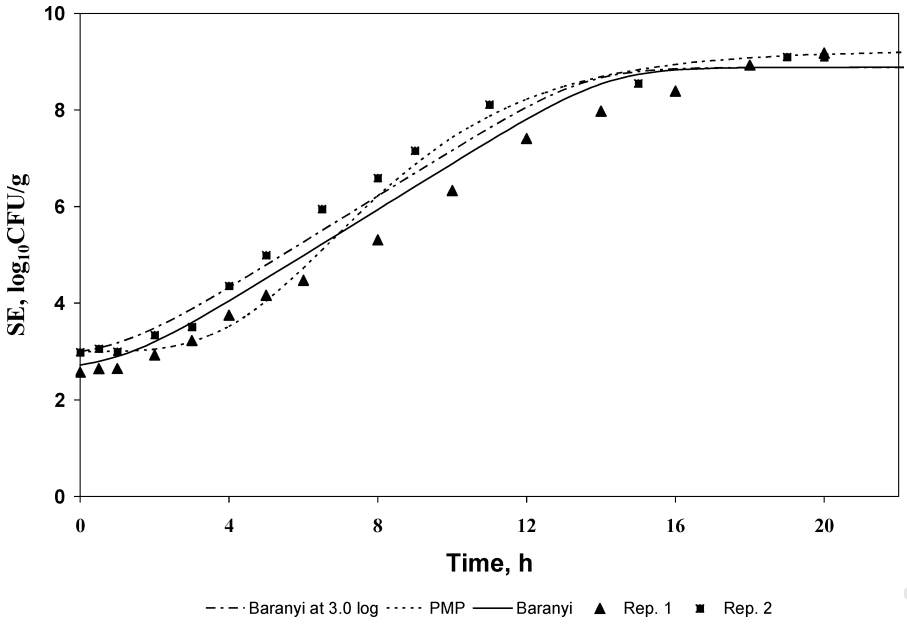

Figure 7—Comparison of the Baranyi model with the PMP model at 30 °C

predicting greater SE populations compared to the observed populations.

A survey of SE in unpasteurized liquid egg revealed that 19% (937 samples) of the U.S. samples were SE positive, with greater SE positives (40%) observed in the Northern region (Hogue and others 1997). As the samples were collected from egg breaking plants, cross contamination due to breaking of shell eggs might have contributed to the higher SE prevalence. When the shell is contaminated externally during laying or during subsequent operations, SE can translocate through the shell, albumen, and eventually into the yolk after yolk membrane (vitelline) break down during storage. The egg yolk membrane acts as a barrier for SE to pass into yolk from albumen. As the permeability of this membrane is temperature dependent, SE invasion of yolk can occur to greater extent at higher temperatures or over during storage at abusive temperatures (Humphrey and others 1994; Grijspeerdt 2005). Grijspeerdt and others (2005) reported that a single SE cell inoculated near the shell surface can translocate, and grow to 5.4 $\log_{10}$ CFU/mL level throughout the albumen and yolk within 78 h, with greater growth observed in the yolk (2 $\log_{10}$ CFU/mL) compared to the albumen.

The dynamic models developed can be used by the industry to evaluate their current egg cooling practices at the collection facility as well as during processing and distribution to minimize the risk of SE growth. Further, the models can be incorporated into the USDA-FSIS risk assessment to improve its performance in evaluating the risk of SE growth in the egg production and shell egg processing and distribution modules. Availability of data on probability of SE occurrences in egg components, a model for migration of SE through egg components (Grijspeerdt and others 2005) and its subsequent growth in yolk predicted from this study can be very useful in filling data gap for risk assessment studies.

Currently, USDA-FSIS requires that the shell eggs be stored and held at an ambient temperature of ≤ 7.2 °C within 12 h of laying and during distribution. Cooling of eggs in the pallet is slow and it can take several days to cool the eggs to 7.2 °C. Maintenance of ambient temperature at less than 7.2 °C may be difficult to achieve during storage because of continuous loading and unloading operations of shell eggs during collection and other mechanical operations. Further, the dynamic SE growth models developed in this study can evaluate the risk of SE growth during cooling of the shell eggs at production and aid in designing safe cooling rates for the eggs, rather than the simplistic recommendation that the eggs be stored at ≤ 7.2 °C within 12 h.

Conclusions

The Baranyi model fitted well the growth data of SE in shell eggs collected at temperatures of 10, 15, 20, 25, 30, 35, 37, 39, 41, and 43 °C covering the entire range for growth and the parameters for the model were estimated at each temperature. SE did not grow at 7 and 45 °C. The estimated μ_{max} values were then fitted into the modified Ratkowsky equation. The RMSE and Pseudo-R^2 values calculated for the primary model at each temperature and for the secondary model suggest that the models fitted well the data well.

A dynamic microbial model was also developed and validated for SE growth in egg yolk. The model performed well when evaluated under various temperature profiles, and can be incorporated into the current PMP software or used independently for prediction of SE in shell eggs either at a given static temperature or at any other continuously varying temperature profile. The predictive model can be used to evaluate risk of SE growth in case of temperature abuse during shell egg cooling, storage, and distribution.

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Queries

Q1 Author: Please check the edit in the sentence 'The Baranyi model…'.

Q2 Author: Please provide the location for manufacturer 'Beckman'.

Q3 Author: The year for reference Amézquita has been changed from 2004 to 2005, as given in the list. Please confirm.

Q4 Author: Please provide the last accessed dates in reference FSIS (2005).